THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 160 cm	320 - 160.
2	**(a)** time period is 1.8 s in both	T = 36/20 = 1.8 and 18/10 = 1.8.
3	**(c)** 10.5 cm	12.5 - 2.
4	**(c)** 850 m	2700 - 1850.
5	**(c)** it is in linear motion and its speed increases	It falls along a straight line, speeding up.
6	**(b)** 8.3 cm	All readings are higher.
7	**(a)** 1200 m	2.5 x 480.
8	**(d)** a rolling ball	Rolling ball.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the conversion does not explain why metre is the SI unit.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Non-standard units give different values for different people, so a common standard unit is needed.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3.6 km = 360000 cm (1); 360000 / 60 = 6000 steps (1).	1+1
12	0.003 km = 3 m (1); 3 m = 300 cm (1).	1+1
13	Swing repeats at equal intervals (1); Leaf does not repeat (1).	1+1
14	The edge is curved (1); Use a thread or tape and then measure it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	1.1 km = 1100 m (1); 1100 / 2.2 (1); 500 turns (1).	1+1+1
16	Use a flexible tape (1); Follow the curve (1); Verify with thread (1).	1+1+1
17	40 / 5 (1); 8 cm (1); Scale used (1).	1+1+1
	OR: Coil length 8 cm = 80 mm (1); 80 / 25 (1); 3.2 mm (1).	1+1+1
18	Rotation: spin about own axis (1); Top, fan (1); Revolution: motion around another point; Earth around Sun, Moon around Earth (1).	1+1+1
	OR: Different reference (1); Passenger moves with the train (1); Observer is fixed to the ground (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Footstep (1). (ii) Their steps are of different sizes (1). (iii) Metre (1). (iv) 9 m = 900 cm (1); a tape gives the same value for everyone because it uses a standard unit (1).	1+1+1+2
20	T1 = 60/30 = 2 s (1); T2 = 30/30 = 1 s (1); Shorter pendulum has smaller period (1); Length ratio 4:1, time ratio 2:1 (1); Conclusion: time period depends on length (1).	1+1+1+1+1
	OR: 1.2 m = 120 cm (1); Student goes 120 cm, snail goes 1.2 cm (1); 120 / 1.2 (1); 100 times (1); Compare in the same unit (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up conversions: 1 m = 100 cm (not 10 cm), 1 cm = 10 mm, 1 km = 1000 m.
- Reading the scale from the side, which gives a wrong value; keep the eye straight above the mark.
- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** each blade moves in a circular path	Blades go round.
2	**(c)** placed touching the object and viewed straight	Reduce gap.
3	**(c)** 0.05 mm	15 mm / 300.
4	**(a)** to and fro motion repeated at equal intervals	Periodic.
5	**(b)** millimetre	1 mm.
6	**(d)** 2 m	About 2 m.
7	**(b)** a falling coin	A fall is not repeated.
8	**(a)** 25000 cm	250 m = 25000 cm.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Repeating after equal time intervals is the meaning of periodic motion.
10	**(c)** A is true but R is false.	A is true: lay the thread along the curve and measure it on a scale. R is false, a thread is flexible.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3 min = 180 s (1); 180/1.5 = 120 (1).	1+1
12	2.4 x 100 = 240 cm (1); 240 x 10 = 2400 mm (1).	1+1
13	Reaction time error becomes small (1); Average time for one oscillation is found by division (1).	1+1
14	Sum = 16.5 (1); 16.5 / 3 = 5.5 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Week: 7 (1); 30 days: 30 (1); One rotation per day (1).	1+1+1
16	Thread bends (1); Rod is rigid (1); Thread then measured straight (1).	1+1+1
17	Pendulum swing: periodic (1); Hands: circular, also periodic (1); Both repeat after equal time (1).	1+1+1
	OR: 80 s total (1); 80 / 40 (1); 2 s (1).	1+1+1
18	12395.1 - 12345.6 (1); 49.5 km (1); Convert to 49500 m (1).	1+1+1
	OR: Rolls along a path (1); Counts turns (1); Gives distance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Ruler (1). (ii) 240 mm (1). (iii) Kilometre (1). (iv) A ruler is straight and cannot bend along a curve (1); stretching changes the length of the thread and gives a wrong value (1).	1+1+1+2
20	Use thread and ruler (1); Record in table (1); Divide C by d for each (1); Values near 3.14 (1); Conclusion: ratio is constant (1).	1+1+1+1+1
	OR: Repeated placing of ruler (1); Reading error (1); Zero error (1); Use a longer tape (1); Take repeated measurements (1).	1+1+1+1+1

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- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 0.0001 m	0.1 mm = 10^-4 m.
2	**(b)** 10 m	More precision.
3	**(d)** 24 hours	A day.
4	**(c)** to show lengths in two different units	Both are used.
5	**(a)** 1 mm	Smallest division.
6	**(c)** 365 days	One year.
7	**(c)** 3 m	5 x 0.6.
8	**(b)** 10.6 cm	11.6 - 1.
9	**(d)** A is false but R is true.	A is false: it is linear motion. R is true.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Reading from the side makes the mark appear shifted; looking straight above avoids it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance = turns x circumference (1); 200 x 1.5 = 300 m (1).	1+1
12	A passenger in a moving train (1); At rest w.r.t. train, moving w.r.t. ground (1).	1+1
13	Use thread along the river (1); Convert using the scale (1).	1+1
14	Earth rotates from west to east (1); Relative motion (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	5 km = 5000 m (1); 5000 / 400 (1); 12.5 laps (1).	1+1+1
16	A sagging tape reads a longer value (1); A twisted tape gives a wrong value (1); The tape should lie along the line being measured (1).	1+1+1
17	2.5 m (1); 4.5 m and 3.6 m (1); Total = 10.6 m (1).	1+1+1
	OR: Convert 1.28 m = 128 cm (1); Difference 0.4 cm (1); Small difference shows that results agree reasonably (1).	1+1+1
18	Eye can be misled (1); Depends on the person (1); Instruments give numbers (1).	1+1+1
	OR: It moves up and down repeatedly (1); Equal time each cycle (1); Until friction stops it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Linear motion (1). (ii) The swing (or the giant wheel) (1). (iii) The butterfly (1). (iv) Yes, it repeats the same turn after every 2 minutes (1), equal time intervals mean periodic motion (1).	1+1+1+2
20	Passenger in a bus (1); At rest wrt seat (1); Moving wrt a tree (1); Moon appears to follow us (1); Observer's frame matters (1).	1+1+1+1+1
	OR: Perimeter of a square = 4 x side (1); One round = 4 x 120 = 480 m (1); Two rounds = 2 x 480 (1); Total = 960 m (1); Distance is measured along the path walked (1).	1+1+1+1+1

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